



SE 328 L: Mobile App Development Lab

CoE – Software Engineering

Alfaisal University

Lab 10: Flutter Part 2

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Objectives

The objective of this lab is to:

The objective of this lab is to develop students' skills in building Flutter applications by creating a chat app and a simple e-commerce app.

Submission Style:

- Add screenshot of the app screen you built in this template then attach it to Moodle - Submit the project as .zip file as well that you saved using Android Studio IDE



Important Note:

Any similar or copied work will automatically receive a grade of zero.

Ex#	Grade	Out of	Grader Comment(s)
1		10	
Total		10	
Signature			

Exercise1: Minimal Chat App • Flutter x Firebase

Answer:

- **Authentication System:** The app integrates **Firebase Authentication** to handle secure user access, supporting standard email and password registration and login.
- **Real-Time Database: Cloud Firestore** serves as the backend, managing user data and chat history with real-time updates powered by Flutter's StreamBuilder widgets.
- **Core Logic:** A dedicated chat service handles privacy by generating unique chat room IDs sorting and merging user UIDs to ensure every conversation remains **private and persistent**.
- **UI Features:** The interface offers a custom chat bubble layout and a **light/dark themetoggle** via the Provider package, along with an auto-scroll feature that keeps the latest messages in view.

Dart codes:

```
import 'package:flutter/material.dart';

class ChatPage extends StatefulWidget {
  final String userName;

  const ChatPage({
    super.key,
    required this.userName,
  });

  @override
  State<ChatPage> createState() => _ChatPageState();
}

class _ChatPageState extends State<ChatPage> {

  final TextEditingController messageController =
    TextEditingController();

  List<Map<String, dynamic>> messages = [

    {
      "text": "Hey! How are you?",
      "isSender": false,
    },

    {
      "text": "I'm good! Working on Flutter lab 😊",
      "isSender": true,
    },
  ],
```

```

    {
      "text": "Same honestly 🤔",
      "isSender": false,
    },
  ];

void sendMessage() {

  if (messageController.text.trim().isEmpty) {
    return;
  }

  setState(() {
    messages.add({
      "text": messageController.text,
      "isSender": true,
    });
  });

  messageController.clear();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: const Color(0xFFFF4F7FB),

    appBar: AppBar(
      backgroundColor: Colors.deepPurple,
      foregroundColor: Colors.white,

      title: Text(widget.userName),
    ),

    body: Column(
      children: [

        Expanded(
          child: ListView.builder(
            padding: const EdgeInsets.all(15),
            itemCount: messages.length,

            itemBuilder: (context, index) {

              final message = messages[index];

              return message["isSender"]
                ? buildSenderMessage(message["text"])
                : buildReceiverMessage(message["text"]);
            },
          ),

        Container(
          padding: const EdgeInsets.all(15),
          color: Colors.white,

```

```

        child: Row(
          children: [

            Expanded(
              child: TextField(
                controller: messageController,

                decoration: InputDecoration(
                  hintText: "Type a message",

                  filled: true,
                  fillColor: Colors.grey.shade100,

                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(15),
                    borderSide: BorderSide.none,
                  ),
                ),
              ),
            ),
          ],
        ),
        const SizedBox(width: 10),
        CircleAvatar(
          backgroundColor: Colors.deepPurple,

          child: IconButton(
            onPressed: sendMessage,

            icon: const Icon(
              Icons.send,
              color: Colors.white,
            ),
          ),
        ),
      ],
    ),
  ],
);
}

Widget buildSendMessage(String text) {
  return Align(
    alignment: Alignment.centerRight,

    child: Container(
      margin: const EdgeInsets.only(bottom: 12),

      padding: const EdgeInsets.all(14),

      decoration: BoxDecoration(
        color: Colors.deepPurple,
        borderRadius: BorderRadius.circular(15),
      ),

      child: Text(

```

```

        text,
        style: const TextStyle(
          color: Colors.white,
        ),
      ),
    ),
  );
}

Widget buildReceiverMessage(String text) {
  return Align(
    alignment: Alignment.centerLeft,

    child: Container(
      margin: const EdgeInsets.only(bottom: 12),

      padding: const EdgeInsets.all(14),

      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(15),
      ),

      child: Text(text),
    ),
  );
}
}

```

```

import 'package:flutter/material.dart';
import 'chat_page.dart';
import 'settings_page.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Connectify"),
      ),

      drawer: Drawer(
        child: Column(
          children: [

            const DrawerHeader(
              child: Center(
                child: Text(
                  "Menu",
                  style: TextStyle(fontSize: 22),

```

```

    ),
  ),
),

ListTile(
  leading: const Icon(Icons.chat),
  title: const Text("Messages"),
  onTap: () {
    Navigator.pop(context);
  },
),

ListTile(
  leading: const Icon(Icons.settings),
  title: const Text("Settings"),

  onTap: () {
    Navigator.push(
      context,
      MaterialPageRoute(
        builder: (context) =>
          const SettingsPage(),
      ),
    );
  },
),

const Spacer(),

ListTile(
  leading: const Icon(Icons.logout),
  title: const Text("Logout"),
  onTap: () {
    Navigator.pop(context);
  },
),
],
),
),

body: ListView(
  padding: const EdgeInsets.all(15),

  children: [

    buildChatCard(context, "Sarah", "Hey!", Icons.person),
    buildChatCard(context, "Ali", "Assignment done?", Icons.person),
    buildChatCard(context, "Mona", "Call me later", Icons.person),
  ],
),
);
}

Widget buildChatCard(
  BuildContext context,
  String name,
  String message,
  IconData icon,

```

```

    ) {
    return GestureDetector(
      onTap: () {
        Navigator.push(
          context,
          MaterialPageRoute(
            builder: (context) => ChatPage(userName: name),
          ),
        );
      },
    );

    child: Container(
      margin: const EdgeInsets.only(bottom: 10),

      padding: const EdgeInsets.all(15),

      decoration: BoxDecoration(
        color: Theme.of(context).cardColor,
        borderRadius: BorderRadius.circular(12),
      ),

      child: Row(
        children: [

          CircleAvatar(child: Icon(icon)),

          const SizedBox(width: 10),

          Column(
            crossAxisAlignment:
              CrossAxisAlignment.start,
            children: [

              Text(name,
                style: const TextStyle(
                  fontWeight: FontWeight.bold),
              ),

              Text(message),
            ],
          ),
        ],
      ),
    ),
  ),
);
}
}

```

```

import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

import 'register_page.dart';
import 'home_page.dart';
import 'theme_provider.dart';

```

```

void main() {
  runApp(

    ChangeNotifierProvider(
      create: (context) => ThemeProvider(),

      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {

    final themeProvider =
      Provider.of<ThemeProvider>(context);

    return MaterialApp(
      debugShowCheckedModeBanner: false,

      theme: themeProvider.currentTheme,

      home: LoginPage(),
    );
  }
}

class LoginPage extends StatelessWidget {
  LoginPage({super.key});

  final TextEditingController emailController =
    TextEditingController();

  final TextEditingController passwordController =
    TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(

      body: Center(
        child: SingleChildScrollView(
          child: Padding(
            padding: const EdgeInsets.all(25),

            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [

                Container(
                  padding: const EdgeInsets.all(20),

                  decoration: BoxDecoration(
                    color: Colors.deepPurple.shade100,
                    shape: BoxShape.circle,

```

```

    ),

    child: const Icon(
      Icons.chat_bubble_outline,
      size: 70,
      color: Colors.deepPurple,
    ),
  ),

  const SizedBox(height: 30),

  const Text(
    "Connectify",
    style: TextStyle(
      fontSize: 32,
      fontWeight: FontWeight.bold,
      color: Colors.deepPurple,
    ),
  ),

  const SizedBox(height: 10),

  Text(
    "Chat smarter with your friends",
    style: TextStyle(
      color: Colors.grey,
      fontSize: 16,
    ),
  ),

  const SizedBox(height: 40),

  TextField(
    controller: emailController,

    decoration: InputDecoration(
      hintText: "Email",

      filled: true,
      fillColor: Colors.white,

      prefixIcon: const Icon(Icons.email_outlined),

      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(15),
        borderSide: BorderSide.none,
      ),
    ),
  ),

  const SizedBox(height: 18),

  TextField(
    controller: passwordController,
    obscureText: true,

    decoration: InputDecoration(
      hintText: "Password",

```

```

        filled: true,
        fillColor: Colors.white,

        prefixIcon: const Icon(Icons.lock_outline),

        border: OutlineInputBorder(
          borderRadius: BorderRadius.circular(15),
          borderSide: BorderSide.none,
        ),
      ),
    ),
  ),

  const SizedBox(height: 30),

  SizedBox(
    width: double.infinity,

    child: ElevatedButton(
      onPressed: () {
        Navigator.push(
          context,
          MaterialPageRoute(
            builder: (context) => const HomePage(),
          ),
        );
      },
      style: ElevatedButton.styleFrom(
        backgroundColor: Colors.deepPurple,
        foregroundColor: Colors.white,

        padding: const EdgeInsets.symmetric(
          vertical: 18,
        ),

        shape: RoundedRectangleBorder(
          borderRadius:
            BorderRadius.circular(14),
        ),
      ),
      child: const Text(
        "Login",
        style: TextStyle(
          fontSize: 16,
          fontWeight: FontWeight.bold,
        ),
      ),
    ),
  ),

  const SizedBox(height: 25),

  GestureDetector(
    onTap: () {
      Navigator.push(
        context,

```

```
MaterialPageRoute(
  builder: (context) =>
    RegisterPage(),
),
);
},

child: const Text(
  "Create new account",
  style: TextStyle(
    color: Colors.deepPurple,
    fontWeight: FontWeight.bold,
  ),
),
),
),
),
],
),
),
),
),
),
);
}
```

```
import 'package:flutter/material.dart';
import 'home_page.dart';

class RegisterPage extends StatelessWidget {
  RegisterPage({super.key});

  final TextEditingController emailController =
    TextEditingController();

  final TextEditingController passwordController =
    TextEditingController();

  final TextEditingController confirmPasswordController =
    TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color(0xFFFF4F7FB),

      appBar: AppBar(
        backgroundColor: Colors.transparent,
        elevation: 0,
      ),

      body: Center(
        child: SingleChildScrollView(
          child: Padding(
            padding: const EdgeInsets.all(25),
```

```
child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [

    Container(
      padding: const EdgeInsets.all(20),

      decoration: BoxDecoration(
        color: Colors.deepPurple.shade100,
        shape: BoxShape.circle,
      ),

      child: const Icon(
        Icons.person_add_alt_1,
        size: 70,
        color: Colors.deepPurple,
      ),
    ),

    const SizedBox(height: 30),

    const Text(
      "Create Account",
      style: TextStyle(
        fontSize: 30,
        fontWeight: FontWeight.bold,
        color: Colors.deepPurple,
      ),
    ),

    const SizedBox(height: 10),

    Text(
      "Join Connectify today",
      style: TextStyle(
        color: Colors.grey.shade700,
      ),
    ),

    const SizedBox(height: 35),

    TextField(
      controller: emailController,

      decoration: InputDecoration(
        hintText: "Email",

        filled: true,
        fillColor: Colors.white,

        prefixIcon: const Icon(Icons.email_outlined),

        border: OutlineInputBorder(
          borderRadius: BorderRadius.circular(15),
          borderSide: BorderSide.none,
        ),
      ),
    ),
  ],
),
```

```

const SizedBox(height: 18),

TextField(
  controller: passwordController,
  obscureText: true,

  decoration: InputDecoration(
    hintText: "Password",

    filled: true,
    fillColor: Colors.white,

    prefixIcon: const Icon(Icons.lock_outline),

    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(15),
      borderSide: BorderSide.none,
    ),
  ),
),

const SizedBox(height: 18),

TextField(
  controller: confirmPasswordController,
  obscureText: true,

  decoration: InputDecoration(
    hintText: "Confirm Password",

    filled: true,
    fillColor: Colors.white,

    prefixIcon: const Icon(Icons.lock_person_outlined),

    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(15),
      borderSide: BorderSide.none,
    ),
  ),
),

const SizedBox(height: 30),

SizedBox(
  width: double.infinity,

  child: ElevatedButton(
    onPressed: () {

      if (passwordController.text !=
        confirmPasswordController.text) {

        showDialog(
          context: context,
          builder: (context) => const AlertDialog(
            title: Text(

```



```

Provider.of<ThemeProvider>(context);

return Scaffold(
  appBar: AppBar(
    title: const Text("Settings"),
  ),
  body: Padding(
    padding: const EdgeInsets.all(20),
    child: Container(
      padding: const EdgeInsets.all(20),
      decoration: BoxDecoration(
        color: Theme.of(context).cardColor,
        borderRadius: BorderRadius.circular(15),
      ),
      child: Row(
        mainAxisAlignment:
        MainAxisAlignment.spaceBetween,

        children: [

          const Text(
            "Dark Mode",
            style: TextStyle(fontSize: 18),
          ),

          Switch(
            value: themeProvider.isDarkMode,

            onChanged: (value) {
              themeProvider.toggleTheme();
            },
          ),
        ],
      ),
    ),
  ),
);
}
}

```

```

import 'package:flutter/material.dart';

class ThemeProvider extends ChangeNotifier {

  bool isDarkMode = false;

  ThemeData get currentTheme =>
    isDarkMode ? darkTheme : lightTheme;
}

```

```
void toggleTheme() {
    isDarkMode = !isDarkMode;
    notifyListeners();
}

final ThemeData lightTheme = ThemeData(
    brightness: Brightness.light,

    scaffoldBackgroundColor: const Color(0xFFFF4F7FB),

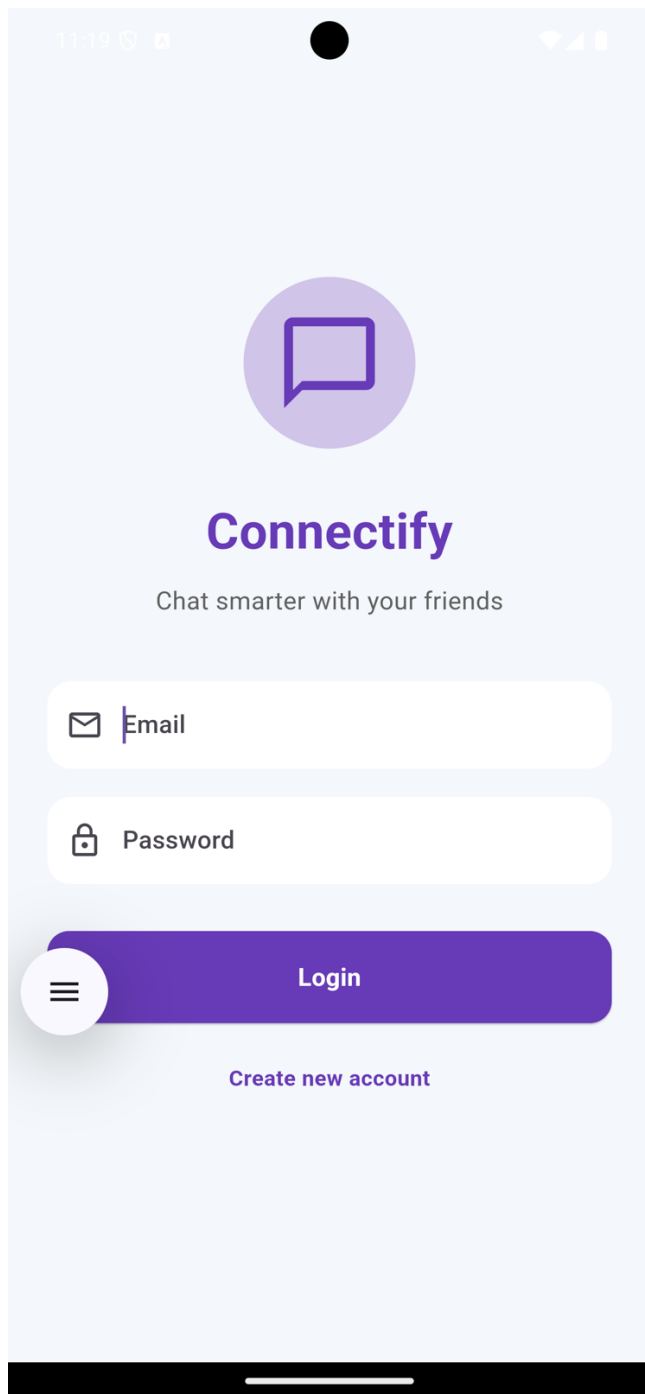
    appBarTheme: const AppBarTheme(
        backgroundColor: Colors.deepPurple,
        foregroundColor: Colors.white,
    ),
);

final ThemeData darkTheme = ThemeData(
    brightness: Brightness.dark,

    scaffoldBackgroundColor: const Color(0xFF121212),

    appBarTheme: const AppBarTheme(
        backgroundColor: Colors.black,
        foregroundColor: Colors.white,
    ),
);
}
```

Screenshots:



11:39



Connectify

Chat smarter with your friends

✉ safatima@alfaisal.edu

🔒|



Login

[Create new account](#)

11:20

Create Account

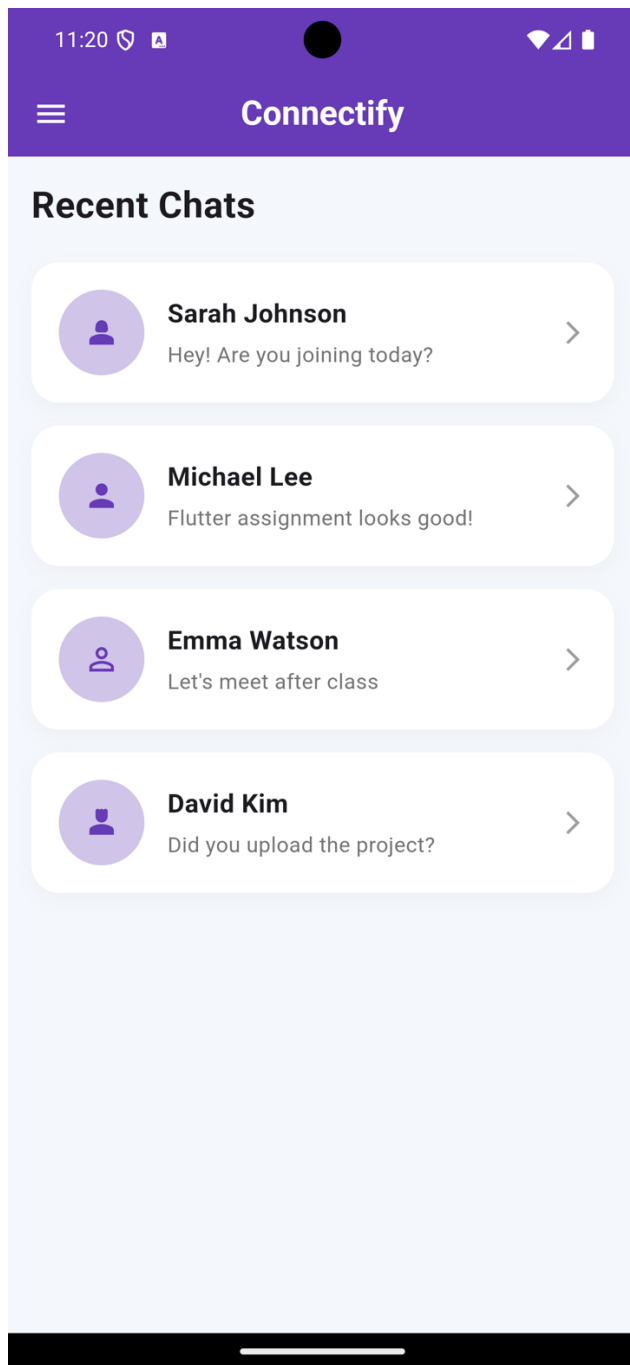
Join Connectify today

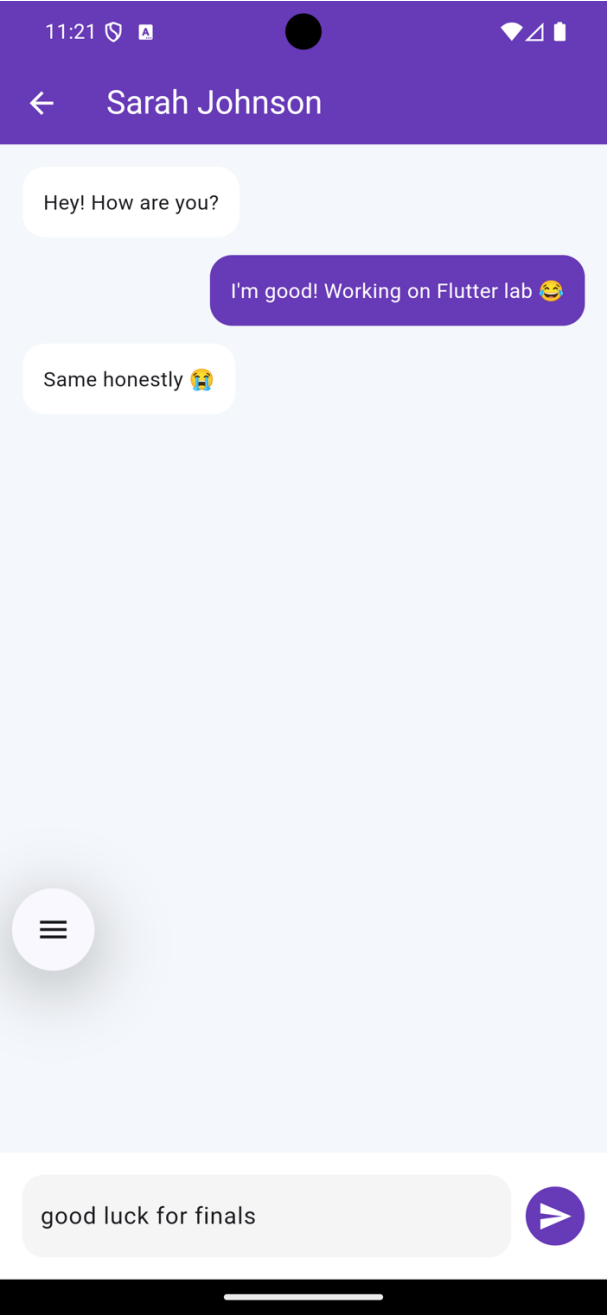
Email

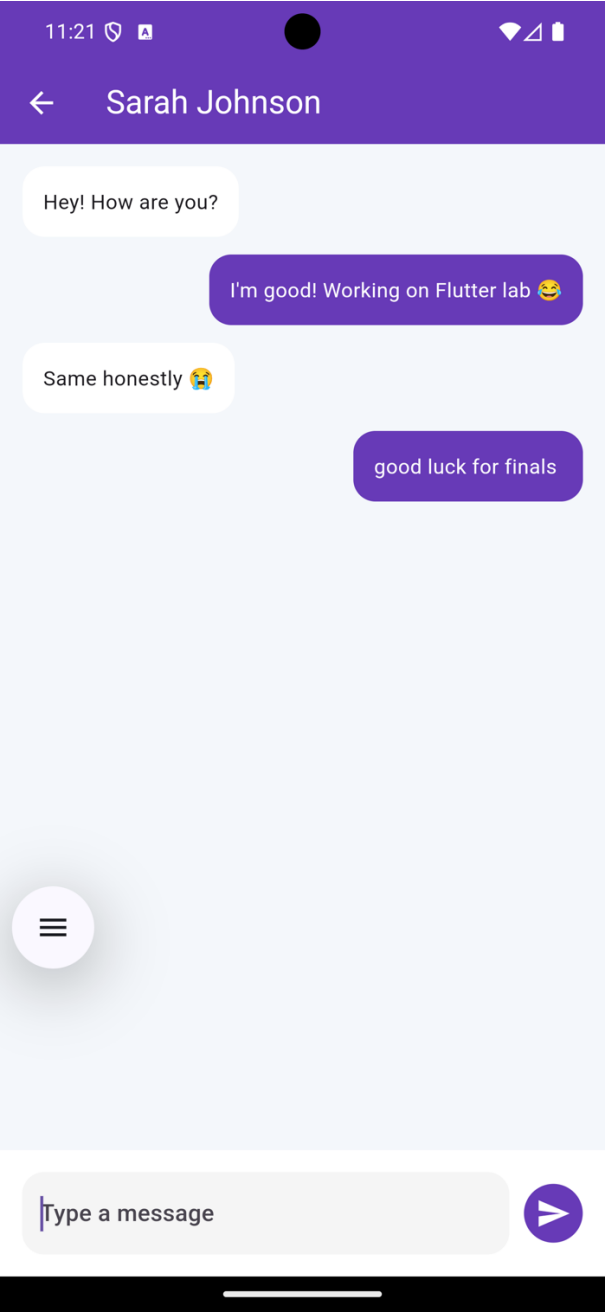
Password

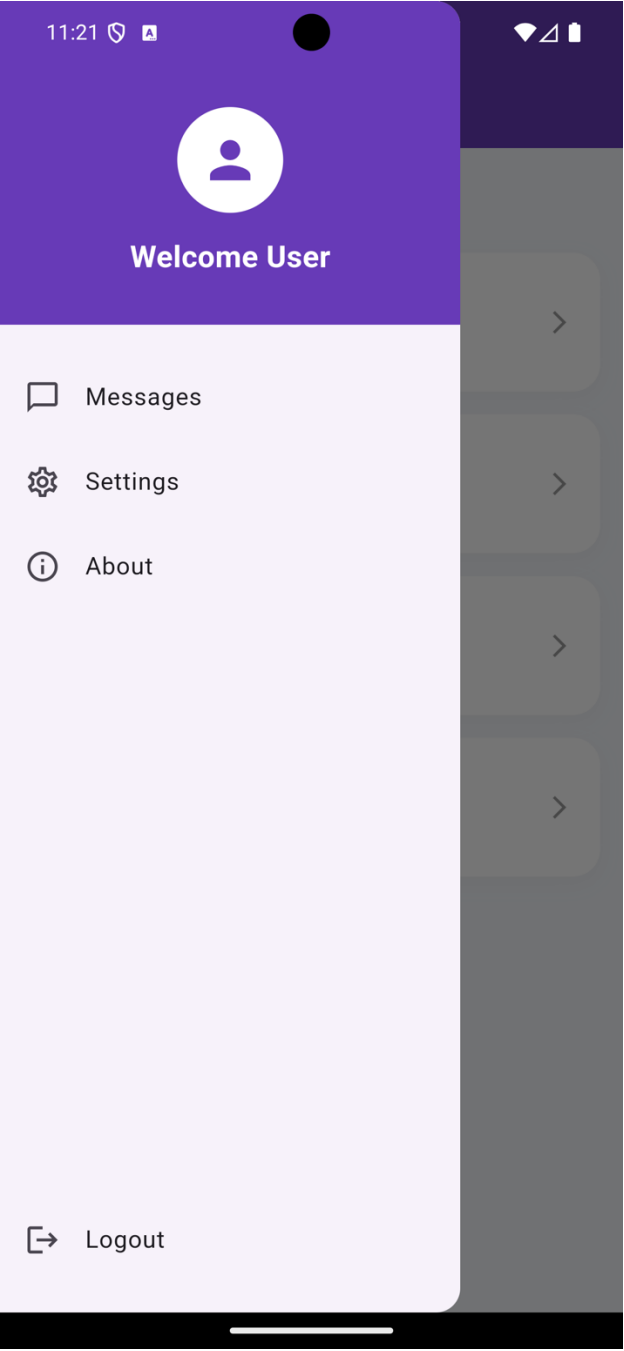
Confirm Password

Register











11:32



Settings

Dark Mode



Exercise1: Minimal Ecommerce App • Flutter

Answer:

- **App Structure:** The application follows a three-screen layout, featuring a welcome landing page, a product catalog for browsing and a dedicated cart page for finalizing selections.
- **State Management:** The **Provider** package handles the cart's logic globally, ensuring that any changes to items are reflected immediately across the entire app.
- **Data Modeling:** Information is organized through a Product class and a Shop model, which collectively manage the inventory list and track the user's active shopping cart.
- **UX Design:** The app prioritizes a minimalist aesthetic, using card style layouts and uniform spacing to clearly present product info and "Add to Cart" actions for a smoother feel.

Dart codes:

my_button.dart:

```
import 'package:flutter/material.dart';
class MyButton extends StatelessWidget {
  final void Function()? onTap;
  final Widget child;
  const MyButton({
    super.key,
    required this.onTap,
    required this.child,
  });
  @override
  Widget build(BuildContext context) {
    return GestureDetector(
      onTap: onTap,
      child: Container(
        decoration: BoxDecoration(
          color: Theme.of(context).colorScheme.primary,
          borderRadius: BorderRadius.circular(12),
        ),
        padding: const EdgeInsets.all(25),
        child: child,
      ),
    );
  }
}
```

```
}
```

my_drawer.dart:

```
import 'package:flutter/material.dart';
import 'my_list_tile.dart';
class MyDrawer extends StatelessWidget {
  const MyDrawer({super.key});
  @override
  Widget build(BuildContext context) {
    return Drawer(
      backgroundColor: Theme.of(context).colorScheme.background,
      child: Column(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children: [
          Column(
            children: [
              // drawer header: logo
              DrawerHeader(
                child: Center(
                  child: Icon(
                    Icons.shopping_bag,
                    size: 72,
                    color: Theme.of(context).colorScheme.inversePrimary,
                  ),
                ),
              ),
              const SizedBox(height: 25),
              // shop tile
              MyListTile(
                text: "Shop",
                icon: Icons.home,
                onTap: () => Navigator.pop(context),
              ),
              MyListTile(
                text: "Cart",
                icon: Icons.shopping_cart,
                onTap: () {
                  Navigator.pop(context);
                  Navigator.pushNamed(context, '/cart_page');
                },
              ),
            ],
          ),
        ],
      ),
    );
  }
}
```

```
),  
],  
),  
// exit tile  
Padding(  
padding: const EdgeInsets.only(bottom: 25.0),  
child: MyListTile(  
text: "Exit",  
icon: Icons.logout,  
onTap: () => Navigator.pushNamedAndRemoveUntil(  
context, '/intro_page', (route) => false),  
),  
),  
],  
),  
);  
}  
}
```

my_list_tile.dart:

```
import 'package:flutter/material.dart';  
class MyListTile extends StatelessWidget {  
  final String text;  
  final IconData icon;  
  final void Function()? onTap;  
  const MyListTile({  
    super.key,  
    required this.text,  
    required this.icon,  
    required this.onTap,  
  });  
  @override  
  Widget build(BuildContext context) {  
    return Padding(  
      padding: const EdgeInsets.only(left: 25.0),  
      child: ListTile(  
        leading: Icon(  
          icon,  
          color: Colors.grey,  
        ),  
        title: Text(  

```

```
text,  
style: TextStyle(color: Theme.of(context).colorScheme.inversePrimary),  
)  
onTap: onTap,  
)  
);  
}  
}
```

my_product_tile.dart:

```
import 'package:flutter/material.dart';  
import '../models/product.dart';  
class MyProductTile extends StatelessWidget {  
  final Product product;  
  final void Function()? onTap;  
  const MyProductTile({  
    super.key,  
    required this.product,  
    required this.onTap,  
  });  
  @override  
  Widget build(BuildContext context) {  
    return Container(  
      decoration: BoxDecoration(  
        color: Theme.of(context).colorScheme.primary,  
        borderRadius: BorderRadius.circular(12),  
      ),  
      margin: const EdgeInsets.all(10),  
      padding: const EdgeInsets.all(25),  
      width: 300,  
      child: Column(  
        crossAxisAlignment: CrossAxisAlignment.start,  
        mainAxisAlignment: MainAxisAlignment.spaceBetween,  
        children: [  
          Column(  
            crossAxisAlignment: CrossAxisAlignment.start,  
            children: [  
              AspectRatio(  
                aspectRatio: 1,  
                child: Container(  
                  decoration: BoxDecoration(  
                    color: Theme.of(context).colorScheme.inversePrimary,  
                    borderRadius: BorderRadius.circular(12),  
                  ),  
                  child: Image.network(  
                    product.imageUrl,  
                    fit: BoxFit.cover,  
                    width: 100%,  
                    height: 100%,  
                  ),  
                ),  
              Text(  
                product.name,  
                style: TextStyle(  
                  color: Theme.of(context).colorScheme.inversePrimary,  
                  fontWeight: FontWeight.bold,  
                ),  
              ),  
            ],  
          ),  
          Text(  
            product.description,  
            style: TextStyle(  
              color: Theme.of(context).colorScheme.inversePrimary,  
            ),  
          ),  
        ],  
      ),  
    ),  
  );  
}
```

```
color: Theme.of(context).colorScheme.secondary,
borderRadius: BorderRadius.circular(12),
),
width: double.infinity,
padding: const EdgeInsets.all(25),
child: Image.asset(product.imagePath),
),
),
const SizedBox(height: 25),
Text(
product.name,
style: const TextStyle(
fontWeight: FontWeight.bold,
fontSize: 20,
),
),
const SizedBox(height: 10),
Text(
product.description,
style: TextStyle(
color: Theme.of(context).colorScheme.inversePrimary,
),
),
],
),
const SizedBox(height: 25),
// product price + add to cart button
Row(
mainAxisAlignment: MainAxisAlignment.spaceBetween,
children: [
Text('\$' + product.price),
Container(
decoration: BoxDecoration(
color: Theme.of(context).colorScheme.secondary,
borderRadius: BorderRadius.circular(12),
),
child: IconButton(
onPressed: onTap,
icon: const Icon(Icons.add),
),
),
],
),
```

```
),  
],  
),  
);  
}  
}
```

product.dart:

```
class Product {  
  final String name;  
  final String price;  
  final String description;  
  final String imagePath;  
  Product({  
    required this.name,  
    required this.price,  
    required this.description,  
    required this.imagePath,  
  });  
}
```

shop.dart:

```
import 'package:flutter/material.dart';  
import 'product.dart';  
class Shop extends ChangeNotifier {  
  // products for sale  
  final List<Product> _shop = [  
    Product(  
      name: "Premium Shoes",  
      price: "99.99",  
      description: "High quality shoes for daily comfort and style.",  
      imagePath: 'assets/images/shoe.jpg',  
    ),  
    Product(  
      name: "Classic Watch",  
      price: "149.99",  
      description: "A timeless timepiece that complements any outfit.",  
      imagePath: 'assets/images/watch.jpg',  
    ),  
    Product(  
      name: "Comfy Hoodie",
```

```

price: "59.99",
description: "Soft and warm hoodie perfect for chilly days.",
imagePath: 'assets/images/hoodie.jpg',
),
Product(
name: "Stylish Glasses",
price: "39.99",
description: "Modern sunglasses to protect your eyes with flair.",
imagePath: 'assets/images/glasses.jpg',
),
];
final List<Product> _cart = [];
List<Product> get shop => _shop;
List<Product> get cart => _cart;
void addToCart(Product item) {
  _cart.add(item);
  notifyListeners();
}
void removeFromCart(Product item) {
  _cart.remove(item);
  notifyListeners();
}
}

```

cart_page.dart:

```

import 'package:flutter/material.dart';
import 'package:minimal_ecommerce_app/components/my_button.dart';
import 'package:provider/provider.dart';
import '../models/product.dart';
import '../models/shop.dart';
class CartPage extends StatelessWidget {
  const CartPage({super.key});
  void removeItemFromCart(BuildContext context, Product product) {
    showDialog(
      context: context,
      builder: (context) => AlertDialog(
        content: const Text("Remove this item from your cart?"),
        actions: [
          MaterialButton(
            onPressed: () => Navigator.pop(context),

```

```
child: const Text("Cancel"),
),
MaterialButton(
  onPressed: () {
    Navigator.pop(context);
    context.read<Shop>().removeFromCart(product);
  },
  child: const Text("Yes"),
),
],
),
);
}

void payButtonPressed(BuildContext context) {
  showDialog(
    context: context,
    builder: (context) => const AlertDialog(
      content:
        Text("User wants to pay! Connect this app to your payment backend"),
    ),
  );
}

@override
Widget build(BuildContext context) {
  // get access to the cart
  final cart = context.watch<Shop>().cart;
  return Scaffold(
    appBar: AppBar(
      backgroundColor: Colors.transparent,
      elevation: 0,
      foregroundColor: Theme.of(context).colorScheme.inversePrimary,
      title: const Text("Cart Page"),
    ),
    backgroundColor: Theme.of(context).colorScheme.background,
    body: Column(
      children: [
        Expanded(
          child: cart.isEmpty
            ? const Center(child: Text("Your cart is empty.."))
            : ListView.builder(
                itemCount: cart.length,
                itemBuilder: (context, index) {
```

```

final item = cart[index];
return Container(
  decoration: BoxDecoration(
    color: Theme.of(context).colorScheme.primary,
    borderRadius: BorderRadius.circular(8),
  ),
  margin: const EdgeInsets.only(left: 25, right: 25, top: 10),
  child: ListTile(
    title: Text(item.name),
    subtitle: Text(item.price),
    trailing: IconButton(
      icon: const Icon(Icons.remove),
      onPressed: () => removeItemFromCart(context, item),
    ),
  ),
);
},
),
// pay button
Padding(
  padding: const EdgeInsets.all(50.0),
  child: MyButton(
    onTap: () => payButtonPressed(context),
    child: const Text("PAY NOW"),
  ),
)
],
),
);
}
}

```

intro_page.dart:

```

import 'package:flutter/material.dart';
import '../components/my_button.dart';
class IntroPage extends StatelessWidget {
  const IntroPage({super.key});
  @override
  Widget build(BuildContext context) {

```

```
return Scaffold(
  backgroundColor: Theme.of(context).colorScheme.background,
  body: Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        // logo
        Icon(
          Icons.shopping_bag,
          size: 72,
          color: Theme.of(context).colorScheme.inversePrimary,
        ),
        const SizedBox(height: 25),
        // title
        const Text(
          "Minimal Shop",
          style: TextStyle(
            fontWeight: FontWeight.bold,
            fontSize: 24,
          ),
        ),
        const SizedBox(height: 10),
        // subtitle
        Text(
          "Premium Quality Products",
          style: TextStyle(
            color: Theme.of(context).colorScheme.inversePrimary,
          ),
        ),
        const SizedBox(height: 25),
        // button
        MyButton(
          onTap: () => Navigator.pushNamed(context, '/shop_page'),
          child: const Icon(Icons.arrow_forward),
        ),
      ],
    ),
  );
}
```

shop_page.dart:

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import '../components/my_drawer.dart';
import '../components/my_product_tile.dart';
import '../models/product.dart';
import '../models/shop.dart';
class ShopPage extends StatelessWidget {
  const ShopPage({super.key});
  void addToCart(BuildContext context, Product product) {
    showDialog(
      context: context,
      builder: (context) => AlertDialog(
        content: const Text("Add this item to your cart?"),
        actions: [
          MaterialButton(
            onPressed: () => Navigator.pop(context),
            child: const Text("Cancel"),
          ),
          MaterialButton(
            onPressed: () {
              Navigator.pop(context);
              context.read<Shop>().addToCart(product);
            },
            child: const Text("Yes"),
          ),
        ],
      ),
    );
  }
  @override
  Widget build(BuildContext context) {
    final products = context.watch<Shop>().shop;
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Colors.transparent,
        elevation: 0,
        foregroundColor: Theme.of(context).colorScheme.inversePrimary,
        title: const Text("Shop Page"),
        actions: [
```

```

IconButton(
  onPressed: () => Navigator.pushNamed(context, '/cart_page'),
  icon: const Icon(Icons.shopping_cart_outlined),
)
],
),
drawer: const MyDrawer(),
backgroundColor: Theme.of(context).colorScheme.background,
body: ListView(
  children: [
    const SizedBox(height: 25),
    // shop subtitle
    Center(
      child: Text(
        "Pick from a selected list of premium products",
        style: TextStyle(
          color: Theme.of(context).colorScheme.inversePrimary),
      ),
    ),
    SizedBox(
      height: 550,
      child: ListView.builder(
        itemCount: products.length,
        scrollDirection: Axis.horizontal,
        padding: const EdgeInsets.all(15),
        itemBuilder: (context, index) {
          final product = products[index];
          return MyProductTile(
            product: product,
            onTap: () => addToCart(context, product),
          );
        },
      ),
    ),
  ],
);
}
}

```

light_mode.dart:

```
import 'package:flutter/material.dart';
ThemeData lightMode = ThemeData(
  colorScheme: ColorScheme.light(
    background: Colors.grey.shade300,
    primary: Colors.grey.shade200,
    secondary: Colors.white,
    inversePrimary: Colors.grey.shade700,
  ),
);
```

main.dart

```
import 'package:flutter/material.dart';
import 'package:minimal_ecommerce_app/themes/light_mode.dart';
import 'package:provider/provider.dart';
import 'models/shop.dart';
import 'pages/intro_page.dart';
import 'pages/cart_page.dart';
import 'pages/shop_page.dart';
void main() {
  runApp(
    ChangeNotifierProvider(
      create: (context) => Shop(),
      child: const MyApp(),
    ),
  );
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const IntroPage(),
      theme: lightMode,
      routes: {
        '/intro_page': (context) => const IntroPage(),
        '/shop_page': (context) => const ShopPage(),
        '/cart_page': (context) => const CartPage(),
      },
    );
  }
}
```

}
}

Screenshots:

